



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

April 21, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 21470
(FOURTH REVISION)

EXPIRATION DATE: 2027-12-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Honeywell Intellectual Properties Inc.
Tempe, AZ
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of non-DOT specification pressure vessel conforming with all regulations applicable to DOT Specification 3HT cylinder, except as specified herein, for the transportation in commerce of the hazardous materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a(a)(1) in that non-DOT specification cylinders are not authorized, except as specified herein.

5. BASIS: This special permit is based on the application of Honeywell Intellectual Properties Inc. dated January 27, 2025, submitted in accordance with § 107.105 and the public proceeding thereon.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Helium, compressed	2.2	UN1046	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Prescribed packagings are non-DOT specification welded spherical pressure vessels made of titanium alloy having nominal water capacity of 1,144, 717, and 259 cubic inches and a maximum service pressure of 10,400 psig. The pressure vessels must be made in accordance with Honeywell drawings 63242357-1 Rev. E, 63242525-1 Rev. E, and 63242304-1 Rev. E, or subsequent revisions on file when these revisions do not alter basic design or performance requirements of the pressure vessel and the pressure vessel continues to meet the design and testing requirements described in the application on file with the Office of Hazardous Materials Safety (OHMS). The pressure vessels must also comply with DOT specification 3HT (§§ 178.35 and 178.44), except as follows:

§ 178.35(e) *Safety devices*. In lieu of the requirements for a safety relief device as prescribed in § 173.302a (a)(2), the pressure vessel is designed with thermal insulation to prevent heat transfer into the pressure vessel in case of fire and has Through Bulkhead Initiators (TBI) installed into the manifold which will auto-ignite at temperature of 450±50 °F and shear the closure fitting for releasing the gas contents of the vessel.

§ 178.44(a) *Type, size, and service pressure*. Welded spherical pressure vessels with nominal water capacity of 1144, 717, and 259 cubic inches and maximum service pressure of 10,400 psig.

§ 178.44(b) *Authorized material.* Titanium alloy (TI-6AL-4V) conforming with AMS4928.

§ 178.44(c) *Identification of Materials.* * * *

§ 178.44(d) *Manufacture.* * * *

§ 178.44(e) *Welding or brazing.* Two forged TI-6AL-4V half spherical shells must be Electron Beam Welded in accordance with AMS2680. All weld joints must be 100 percent radiographed prior to hydrotest and dye-penetrant inspected before and after hydrostatic test.

§ 178.44(f) *Wall thickness.*

(1) * * *

(2) Stresses must be calculated using closed-form hoop stress equations appropriate for thin-walled spherical vessel as submitted to, and on file, with the OHMS.

(3) Not applicable.

§ 178.44(g) *Heat treatment.* All pressure vessels (forged half shells and welded assembly) must be uniformly and properly heat-treated (multiple solution heat treat and aging treatments in accordance with AMS2801, Temper STA1000 – No Alpha case permitted) and inspected by liquid penetrant (ASTM E1417/E1417M/MIL-STD-1907) and ultrasonic (AMS-STD-2154) methods as applicable, in accordance with details on the Honeywell drawings referenced in paragraph 7.a., and on file with the OHMS.

§ 178.44(h) *Openings in cylinders and connections (valves, fuse plugs, etc.) for those openings.* * * *

§ 178.44(i) *Pressure testing.* * * *

Each cylinder must be hydrostatically tested to minimum of 5/3 times the service pressure.

§ 178.44(j) *Cycling test.* Prior to initial shipment of any specific pressure vessel design, cyclic pressurization tests must be performed on at least three representative samples without failure as follows:

(1) Pressurization must be performed hydrostatically between approximate zero and the service pressure at a rate not more than 10 cycles per minute. Adequate recording instrumentation must be provided to document the test cycles. The pressure vessels shall then be subjected to a burst test.

(2) Cycle testing not required on production lots.

(3) In this special permit the “lot” means a group of pressure vessels successively produced having the same size and configuration, same specified material of construction and material heat, manufactured by the same process and heat treated in the same equipment under the same conditions of time, temperature, and atmosphere. In no case may the lot size exceed 200 cylinders; however, any pressure vessel processed for use in the required destructive testing need not be counted as being one of the 200.

(4) * * *

§ 178.44(k) *Burst test*. Burst test shall be performed on design qualification cycle test units and one unit taken at random from each production lot shall be hydrostatically tested to destruction.

§ 178.44(l) *Flattening test*. Not required.

§ 178.44(m) *Physical test*. A physical test must be conducted to determine yield strength, tensile strength, elongation, and reduction of area of material, as follows:

(1) The pressure vessel forgings shall be 100% tested using a minimum of two tensile specimens extracted from a prolongation attached to each forging which is heat treat cycled with the production pressure vessels. The coupons shall conform dimensionally to MTL1106-1.

(2) The physical properties shall conform to those specified on the Honeywell drawings referenced in paragraph 7a above and listed under acceptable results in § 178.44(p).

(3) Metallographic, microstructure and fracture toughness tests are one-time tests at design qualification prior to entering production and must be in accordance with the application on file with the OHMS.

§ 178.44(n) *Magnetic particle inspection*. Not required. Instead, each pressure vessel must be inspected by a qualified inspector using apparatus and procedures for liquid penetrant examination in accordance with MIL-STD-6866, with

acceptance criteria in accordance with MIL-STD-1907. Inspection must be performed externally on the finished pressure vessel after hydrostatic test. Evidence of discontinuities, which in the opinion of the inspector may appreciably weaken or decrease the durability of the pressure vessel, must be cause for rejection. DOT approved Independent Inspection Agency (IIA) shall have oversight for all design qualification and production inspections and test records.

§ 178.44(o) *Leakage test*. Not applicable.

§ 178.44(p) *Acceptable results of tests*. Results of the physical tests, burst test, and cycling test must conform to the following:

- (1) Flattening: not required.
- (2) Physical tests. The following minimum mechanical properties must be met:
 - (i) Ultimate tensile strength = 155,000 psi
 - (ii) Yield strength = 140,000 psi
 - (iii) Elongation = 10% Min on 4D gauge length (Transverse specimen)
 - (iv) Reduction in area = 20% Min
 - (v) Fracture toughness (at design qualification), $K1C \geq 42 \text{ ksi-in}^{0.5}$
- (3) The burst pressure must be at least $\frac{4}{3}$ times the test pressure.
- (4) Cycling: the test pressure vessels must withstand at least 1000 pressurization cycles without any evidence of distortion or failure.
- (5) The entire production lot shall be rejected if any of the physical tests fail to meet the specified requirements.

§ 178.44(q) *Rejected pressure vessels*. A pressure vessel lot that was rejected may be repaired and requalified as follows:

- (1) Reheat treatment is authorized; subsequent thereto, acceptable pressure vessels must pass all prescribed tests.

(2) Repair of welds is authorized after which reheat treatment is required and, subsequent thereto, acceptable pressure vessels must pass all prescribed tests.

(3) If a pressure vessel lot is rejected and the cause for the rejection is determined by test or inspection, and the defective pressure vessels are eliminated, then the remaining pressure vessels may be qualified as a new lot.

§ 178.44(r) *Marking.*

(1) Applies, except that:

(i) Instead of DOT-3HT, pressure vessels must be marked “DOT-SP 21470” followed by the service pressure.

(ii) Marking must be by low stress type stamping method such as electro-chemical etching, vibro-pen or laser marking which does not decrease the integrity of the pressure vessel; and in an area, and to a depth, which will ensure that the wall thickness measured from the root of the stamping to the interior surface is equal to or greater than the minimum prescribed wall thickness. Marking must be permanent and legible.

(iii) Marking must be permanent and legible and must include the date of manufacture.

(2) Stamping of elastic expansion is not required.

(3) * * *

§ 178.44(s) *Inspector's report.* * * *

The inspector's report may be appropriately modified to reflect identification and conformance with this special permit.

b. REQUALIFICATION: Not required. The pressure vessels are non-refillable, for single use only.

8. SPECIAL PROVISIONS:

a. Prior to first shipment of pressure vessels under the terms of this special permit, the qualification tests specified in this special permit must be completed with satisfactory results. The test results must be submitted to the OHMS and be acknowledged in writing.

- b. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.
 - c. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.
 - d. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - e. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.
 - f. A current copy of this special permit must be maintained at each facility where the packaging is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - g. Pressure vessels must be shipped in strong non-bulk outer packagings in conformance with § 173.301(a)(9).
 - h. The pressure vessels are for single discharge use only and are not authorized 40 years after the date of manufacture.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each motor vehicle used to transport packages covered by this special permit.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) — “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: ae